DEPARTMENT OF MECHANICAL ENGINEERING
RICE UNIVERSITY

MECH/ELEC/COMP 498/598: INTRODUCTION TO ROBOTICS

Project: Let's Start a Robot Company

Project Purpose:

The goal of this project is to reinforce the concepts learned in the course labs through an **individual, creative application**.

Each student will create their own **fictional startup company** whose mission is to raise venture funding to build a robot that solves a real-world (or space-based) problem. As part of this startup, you will design your own **robot manipulator** and apply the same core concepts used in the labs to this new robot.

To reduce overhead, **starter code will be provided** that mirrors the structure of the lab assignments. In most cases, you will be able to **reuse and adapt the functions you wrote for the labs**, modifying them as needed to work with your unique robot design rather than starting from scratch.

This repetition is intentional: applying the same ideas to a new system—of your own design—is one of the most effective ways to solidify understanding and build confidence with the material.

Project Structure and Relationship to Labs:
The project will mirror the labs almost exactly:

LAB	*PROJECT COMPONENT*
1	Forward Kinematics
2	Inverse Kinematics
3	Robot Trajectory Creation
4	Dynamics (required for grads, options for undergrads)

Each student will:
- Define a custom robot for their startup
- Implement (an often simpler versions of) the lab methods for that robot
- Submit their code through GradeScope (some portions will be auto-graded)

If you complete the project **immediately after finishing the corresponding lab**, the additional workload should be modest.
If you participated actively in your lab team, the project portion should take you less than half the time the lab did.

Proposal (5 Points)

Purpose: Lock in your idea early, avoid surprises, and set yourself up to execute the project parallel to the labs

You will submit a short proposal (1/2 page) describing your startup and the robot you intend to make for that startup.
All on-time submissions receive full credit, regardless of content.
If the instructor has a concern about your robot, they will reach out.

Proposal must include:
- Student name
- Company name
- Problem the company is solving
- High-level description of the robot

Robot Technical Deliverables (80 Points)

1. **Robot Design and Forward Kinematics (20 Points)**
   Robot Complexity Requirements:
   - Undergraduates: minimum **4** degrees of freedom
   - Graduate students: minimum **5** degrees of freedom

   Deliverables:
   (10 points) Graphic of the robot with DH parameters
   (10 points) Forward kinematics implementation (auto-graded)

2. **Inverse Kinematics (20 points):**
   All student must implement the inverse kinematics for their robot
   - Code submitted through GradeScope
   - Fully auto-graded

3. **Robot Demonstration / Visualization (20 points)**
   You will create a simulated motion visualization showing your robot performing one or more tasks relevant to your startup.
   - Submitted as code through GradeScope
   - Manually graded by the instructor
   - If you simply visualize the robot similar to lab 3, that will receive 15 points. Adding more creativity or content to the visualization will earn the full 20 points.

   Note: The visualization can (and should) be used in the pitch deck.

4. **Dynamics or Mechanical Design (20 points)**
   Graduate students – Dynamics is required
   Undergraduates – Option between Dynamics or Mechanical Design

   Option A: Dynamics:
   Implement robot dynamics and demonstrate:
   1. *Passive Motion*
      - Robot initialized in an unstable configuration
      - Motion under gravity (or applied forces)
      - Plots of:
        - Potential Energy
        - Kinetic Energy
        - Total Energy (should be conserved)
   2. *Controlled Motion*
      - Simple P / PD / PID controller
      - Robot performs its task
      - Plots of:
        - Joint torques
        - Joint velocities
      - Values must be physically reasonable for your robot

   Option B: Mechanical Design
   Students must submit a sufficiently detailed design of their robot, the end effector, or a joint. This design should show sufficient detail to clearly demonstrate how it would work. Given the amount of the robot that you design dictates the level of detail required.
   - Full Robot – Detailed shape of the arm, location of actuators (modeled as realistic cylinders with a selected motor and sensor) and a description of the end effector.
   - End-Effector – Detailed design (in CAD) of the end effector linkages with descriptions of actuation
   - Joint – Detailed design (in CAD) of the joint including the motor, sensors, and bearings.

Pitch Deck (15 Points + Bonus)

You will create a startup pitch deck as if presenting to early-stage venture investors.

Required Content:

- The market and the problem you are solving
- Why that problem is worth solving
- Your robotic solution
- Why *you* are uniquely positioned to build this company

Expectations:

- This pitch is about story telling, not an academic presentation
- Therefore, it should include more figures and images than words or equations
- This pitch deck is a creative outlet to sell your company and a fun conclusion to demonstrate the technical work you put into the lab.

Bonus:

The top 3 pitch decks as judged by "most likely to be funded" will be given 5 bonus points each.

Final Notes:

- The autograder will be available before Lab1 is due so you may start submitting portions of your project.
- You may request up to three (3) early grades of the non-auto-graded components (in part or in whole) up to 2 weeks before the final deadline.
- The project is designed to reward steady, incremental progress.